



Check for updates

SYSTEMATIC REVIEW

Artificial Intelligence Adoption as a Driver of Innovation and Competitiveness in SMEs: A Bibliometric and Systematic Review

[version 1; peer review: 2 approved]

Yohannes Yesuf , Ziska Fields

Business Management, University of Johannesburg College of Business and Economics, Auckland Park, Gauteng, South Africa

V1 First published: 31 Oct 2025, 14:1187
<https://doi.org/10.12688/f1000research.171494.1>
Latest published: 31 Oct 2025, 14:1187
<https://doi.org/10.12688/f1000research.171494.1>

Abstract

This study employs a bibliometric and systematic literature review (B-SLR) approach, guided by the PRISMA 2021 framework, to investigate the transformative impact of Artificial Intelligence (AI) on small and medium-sized enterprises (SMEs). Drawing on peer-reviewed studies published between 2020 and 2024 across the Scopus databases, the review integrates quantitative mapping and qualitative synthesis to identify emerging research patterns, thematic concentrations, and methodological trends. The findings reveal a marked surge in scholarly engagement since 2020, reflecting growing recognition of AI as both a strategic and operational driver of competitiveness in SMEs. Three dominant thematic clusters emerge: (1) AI-driven innovation, encompassing applications such as machine learning, predictive analytics, automation, and process optimization; (2) entrepreneurial decision-making, highlighting how AI supports opportunity recognition, strategic planning, and market sensing; and (3) adoption barriers and enablers, focusing on organizational readiness, digital infrastructure, human capital, and financial capability. While the field demonstrates expanding theoretical and practical interest, significant gaps remain regarding methodological diversity, contextual variation across developing economies, and the integration of socio-ethical considerations in AI adoption. The study contributes by consolidating fragmented knowledge, mapping the intellectual structure of the domain, and offering actionable implications for policymakers, practitioners, and scholars seeking to leverage AI to foster innovation, digital resilience, and sustainable growth in SMEs.

Open Peer Review

Approval Status

	1	2
version 1		
31 Oct 2025	view	view

1. **Erstu kassa**, Woldia University, Weldiya, Ethiopia
2. **Arjita Jain** , NCRD's Sterling Institute of Management Studies,, Navi Mumbai, India

Any reports and responses or comments on the article can be found at the end of the article.

Keywords

Adoption, artificial intelligence, entrepreneurial decision-making, innovation, small and medium-sized enterprises



This article is included in the **Artificial Intelligence and Machine Learning** gateway.

Corresponding author: Yohannes Yesuf (makyohannes1@gmail.com)

Author roles: **Yesuf Y:** Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Writing – Original Draft Preparation, Writing – Review & Editing; **Fields Z:** Project Administration, Resources, Supervision, Validation, Writing – Review & Editing

Competing interests: No competing interests were disclosed.

Grant information: The author(s) declared that no grants were involved in supporting this work.

Copyright: © 2025 Yesuf Y and Fields Z. This is an open access article distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The author(s) is/are employees of the US Government and therefore domestic copyright protection in USA does not apply to this work. The work may be protected under the copyright laws of other jurisdictions when used in those jurisdictions.

How to cite this article: Yesuf Y and Fields Z. **Artificial Intelligence Adoption as a Driver of Innovation and Competitiveness in SMEs: A Bibliometric and Systematic Review [version 1; peer review: 2 approved]** F1000Research 2025, **14**:1187 <https://doi.org/10.12688/f1000research.171494.1>

First published: 31 Oct 2025, **14**:1187 <https://doi.org/10.12688/f1000research.171494.1>

Introduction

Small and Medium-sized Enterprises (SMEs) constitute the backbone of global economies, serving as key contributors to innovation, employment generation, and sustainable growth across advanced and emerging markets (Ndubisi et al., 2021). Accounting for nearly 90% of global businesses and more than half of total employment, SMEs are indispensable in fostering entrepreneurship and driving socio-economic dynamism (World Bank Group, 2022). Beyond their economic significance, SMEs are vital engines of innovation, often experimenting with novel products, services, and business models to respond to rapidly shifting customer demands and competitive pressures (Albats et al., 2023). However, their innovation capacity is frequently constrained by structural and operational challenges, such as limited financial resources, weak technological capabilities, managerial gaps, and inadequate access to advanced digital infrastructures (Wang et al., 2025). These barriers hinder their ability to achieve competitiveness in increasingly digital and knowledge-driven economies.

The advent of Artificial Intelligence (AI) presents a transformative pathway for SMEs to address these limitations and enhance their competitiveness (Iyelolu et al., 2024). AI technologies, including machine learning, natural language processing, computer vision, and predictive analytics, equip firms with tools to automate processes, improve decision-making, and generate innovative solutions that strengthen both operational efficiency and market responsiveness (Ghobakhloo et al., 2023). In the SME context, AI adoption has been shown to accelerate product development, deepen customer engagement, strengthen market intelligence, and improve strategic decision-making (Iyelolu et al., 2024). Moreover, AI enables the recognition of opportunities and evidence-based entrepreneurial choices, making it a powerful driver of innovation and competitiveness, especially in uncertain and resource-constrained environments (Giuggioli & Pellegrini, 2023).

Nevertheless, AI adoption within SMEs is neither uniform nor straightforward. While some enterprises have successfully embedded AI into their operations, many remain at the experimental or exploratory stage, largely due to organizational, technological, and institutional barriers (Khan et al., 2025). Factors such as digital readiness, access to financial and human capital, and strategic alignment are critical in determining the success of adoption initiatives (Omowole et al., 2024). Similarly, external pressures, including regulatory environments, market competition, and access to digital ecosystems, strongly shape adoption trajectories (Zahra et al., 2023). This heterogeneity underscores the need for a systematic examination of both the drivers and barriers to AI adoption across diverse contexts.

Although scholarly attention to AI adoption in SMEs has expanded, research remains fragmented and unevenly distributed across disciplinary and geographical boundaries (Ayinaddis, 2025; Siddiqui et al., 2024). Much of the existing work focuses on technical implementations or single-case explorations, providing limited insights into broader theoretical, methodological, and practical patterns (Schwaake et al., 2025). Furthermore, most studies disproportionately emphasize developed economies, leaving SMEs in emerging economies, where challenges and opportunities are distinct, relatively underexplored (OECD, 2024). This imbalance restricts the generalizability of findings and limits the development of context-sensitive strategies for the adoption of AI.

To bridge these gaps, this study integrates bibliometric and systematic literature review (B-SLR) approaches to provide a comprehensive overview of the intellectual landscape on AI adoption in SMEs. Bibliometric analysis enables the identification of key research trends, influential authors, collaboration networks, and thematic clusters (Hassan & Duarte, 2024), while the SLR approach synthesizes empirical evidence, highlights methodological tendencies, and identifies critical barriers, enablers, and innovation outcomes (Marzi et al., 2025). Together, these methods provide a dual lens for mapping the evolution of the field and advancing theoretical and practical insights. This study specifically addresses the question: How has the adoption of AI in SMEs evolved over time? What dominant themes and methodologies characterize the field? What factors drive or hinder AI adoption, and how do these shape innovation and competitiveness outcomes?

In doing so, this research makes three contributions. First, it enriches academic understanding by systematically mapping the technological, organizational, and environmental determinants of AI adoption in SMEs and linking them to outcomes of innovation and competitiveness. Second, it provides practitioners with actionable insights on how AI can be leveraged to enhance process efficiency, product innovation, and market positioning. Third, it provides policymakers with evidence-based guidance on designing supportive infrastructures, funding mechanisms, and regulatory frameworks that facilitate inclusive AI-driven growth, particularly in resource-constrained and emerging contexts. Ultimately, the study advances the discourse on AI adoption in SMEs by integrating fragmented scholarship, uncovering research gaps, and setting an agenda for future studies at the intersection of AI, entrepreneurship, and innovation.

AI Adoption, innovation, and competitiveness

AI adoption is increasingly recognized as a catalyst for innovation and competitiveness in SMEs. However, recent studies caution that the benefits are not automatic and largely depend on organization-specific and contextual conditions. Large-scale surveys and empirical evidence indicate a rapid increase in the use of generative and applied AI across businesses; however, the extent of adoption and its outcomes vary considerably depending on organizational capabilities, sectoral dynamics, and institutional environments (Arroyabe et al., 2024). The literature highlights three key domains where AI adoption can enhance SME competitiveness: operational efficiency, innovation in products and services, and responsiveness to market demands (Carayannis et al., 2025; Iyelolu et al., 2024). Process automation and data-driven analytics help reduce costs, streamline production cycles, and enhance decision-making (Adekunle et al., 2021). Sector-based evidence also indicates notable improvements in productivity and export performance among firms with higher digital maturity and more advanced AI integration (Segarra-Blasco et al., 2025). Despite this, many SMEs still achieve only modest gains because AI adoption is often limited to narrow applications, such as customer support, scheduling, or marketing, rather than being embedded within broader business model transformation and innovation strategies (Castillo-Vergara et al., 2025).

To explain these differences, scholars often employ the Technology-Organization-Environment (TOE) framework in conjunction with the dynamic capabilities perspective. From the TOE perspective, technological readiness (e.g., data availability and platforms), organizational preparedness (e.g., leadership commitment and employee skills), and external conditions (e.g., regulatory support and competitive pressure) jointly shape both adoption and its outcomes (Sánchez et al., 2025). The dynamic capabilities approach further emphasizes absorptive capacity, learning orientation, and agility as mediators in the process of translating AI investment into innovation and competitive advantage (Ferreira et al., 2021).

Several studies argue that complementary investments are essential for AI adoption to deliver transformative results. Cross-country analyses reveal that SMEs achieve strategic benefits from AI only when investments are made in workforce reskilling, data governance, managerial experimentation, and integration into core business processes (Arroyabe et al., 2024). Without these, AI tends to generate localized productivity improvements that fail to translate into long-term competitiveness. At the same time, persistent barriers continue to limit meaningful AI adoption among SMEs. Financial and human resource constraints, inadequate digital infrastructure, poor data quality, and insufficient managerial knowledge are frequently reported challenges (e.g., Omowole et al., 2024; Omrani et al., 2022). Industry reports also note that many SMEs do not invest adequately in in-house training, resulting in adoption being concentrated in firms with stronger financial capacity or international linkages (Nazir et al., 2025).

Ecosystem-level interventions are highlighted as critical in bridging these gaps. European policy analyses indicate that public initiatives, sector-specific incubators, affordable access to cloud platforms, and targeted skills development programs enhance adoption rates and facilitate a shift from basic AI use to innovation-driven integration (Schneegans et al., 2021). Such initiatives are particularly vital in low-resource settings, where market limitations and coordination failures constrain organizations' capacity to invest in digital capabilities. Nonetheless, emerging critical perspectives caution against overly optimistic assumptions. Recent industry surveys indicate that many generative AI pilots underperform, primarily due to a lack of alignment with organizational strategy and inadequate governance structures (Feher, 2025). This reinforces the argument that strong leadership, effective change management, and robust governance mechanisms are essential in transforming experimental AI adoption into a sustainable competitive advantage (Kulkov et al., 2024).

Overall, the latest scholarship underscores that while AI can serve as a powerful enabler of innovation and competitiveness for SMEs, its success depends on complementary capabilities within the organizations, supportive institutional ecosystems, and deliberate strategic leadership. Policy measures that enhance digital skills, strengthen data infrastructure, and provide tailored implementation support appear particularly effective in enabling SMEs to translate AI adoption into durable innovation outcomes.

Methodology

This study adopts a dual methodological approach, B-SLR, to investigate AI adoption in SMEs. The rationale for this combined approach lies in the complementary strengths of the two methods: bibliometric analysis provides a quantitative, macro-level understanding of research trends, influential authors, collaboration networks, and thematic structures, whereas SLR offers a qualitative, in-depth synthesis of empirical findings, methodological approaches, and theoretical contributions (Aria & Cuccurullo, 2017; Marzi et al., 2025). Integrating these methods enhances the validity, comprehensiveness, and interpretive richness of the study, allowing it to capture both the breadth and depth of scholarship on AI adoption in SMEs. Such a dual approach is particularly justified in fragmented research domains, where studies are dispersed across disciplines like management and entrepreneurship, and where insights from one method alone may provide an incomplete picture (Marzi et al., 2025).

Bibliometric analysis

Bibliometric analysis is widely recognized for its rigor in systematically mapping scientific knowledge and identifying intellectual structures within a research field (Donthu et al., 2021). In this study, publications were retrieved from Scopus, selected for their extensive coverage, high indexing quality, and broad representation of peer-reviewed scholarly outputs. The keyword strategy, incorporating terms such as “Artificial Intelligence,” “AI adoption,” “SMEs,” “small and medium enterprises,” and “entrepreneurship,” was designed to capture literature reflecting both technological applications and managerial implications. The temporal scope of 2020-2024 was chosen to encompass the evolution of AI research from early conceptual explorations to recent developments influenced by Industry 4.0, digital transformation, and emerging AI-driven business models.

After cleaning the dataset to remove duplicates and irrelevant records, a bibliometric analysis was conducted using VOSviewer. The software constructs and visualizes networks of co-authorship, co-citation, and keyword co-occurrence, offering clear representations of intellectual structures, thematic clusters, and research frontiers. This approach provides systematic and replicable insights into influential scholars, institutions, emerging themes, and underexplored areas in the adoption of AI by SMEs. The use of bibliometric techniques is therefore justified, as they enable evidence-based identification of research frontiers, knowledge gaps, and intellectual structures, ensuring a comprehensive overview of this rapidly expanding field (Kraus et al., 2024).

Systematic Literature Review (SLR)

The SLR complements the bibliometric mapping by providing qualitative depth, synthesizing empirical findings, methodologies, and theoretical perspectives related to AI adoption in SMEs. The SLR followed PRISMA guidelines (Page et al., 2021), ensuring transparency, reproducibility, and methodological rigor. This approach is particularly justified given the fragmented nature of AI adoption research, which spans multiple disciplines, sectors, and geographies. By systematically evaluating existing studies, the SLR helps identify factors influencing adoption, innovation outcomes, and barriers/enablers, while providing a theoretical and practical lens on managerial decision-making and competitiveness.

The review process followed a structured multi-stage approach to ensure rigor and reliability. In the identification stage, a total of 177 peer-reviewed journal articles published in English were retrieved from Scopus using the same keyword strategy applied in the bibliometric analysis. Grey literature, conference proceedings, and non-peer-reviewed sources were excluded to maintain the research’s quality and integrity. During the screening stage, titles and abstracts were reviewed, resulting in the retention of 89 studies that empirically examined AI adoption in SMEs with a focus on organizational, managerial, or entrepreneurial aspects. In the eligibility assessment, full texts were assessed against predefined inclusion criteria, and studies concentrating exclusively on large organizations, purely technical AI development, or non-SME contexts were excluded. This refinement justified the focus on SMEs as distinct organizational entities with unique challenges, leaving 63 eligible articles for analysis (see Table 1 and Figure 1). Finally, in the data extraction stage, structured information was captured from each study, including publication year, regional and sectoral context, research objectives and questions, methodological approaches (qualitative, quantitative, or mixed methods), AI focus (technology type and application domain), and key findings related to innovation, entrepreneurial decision-making, and competitiveness outcomes.

Thematic synthesis was employed to code the extracted data and identify key themes, including AI-driven innovation, adoption enablers and barriers, and managerial strategies. Combining quantitative bibliometric mapping with qualitative

Table 1. Overview of the B-SLR identification and screening process.

Phase	Description	Records (n)
Identification	Records identified through Scopus using a keyword strategy	177
	Records after duplicates removed	116
Screening	Records screened by title and abstract	89
	Records excluded (e.g., not related to SMEs, irrelevant scope)	67
Eligibility	Full-text articles assessed for eligibility	89
	Full-text articles excluded (e.g., large organizations only, purely technical focus, not empirical)	26
Included	Studies included in the final synthesis	63

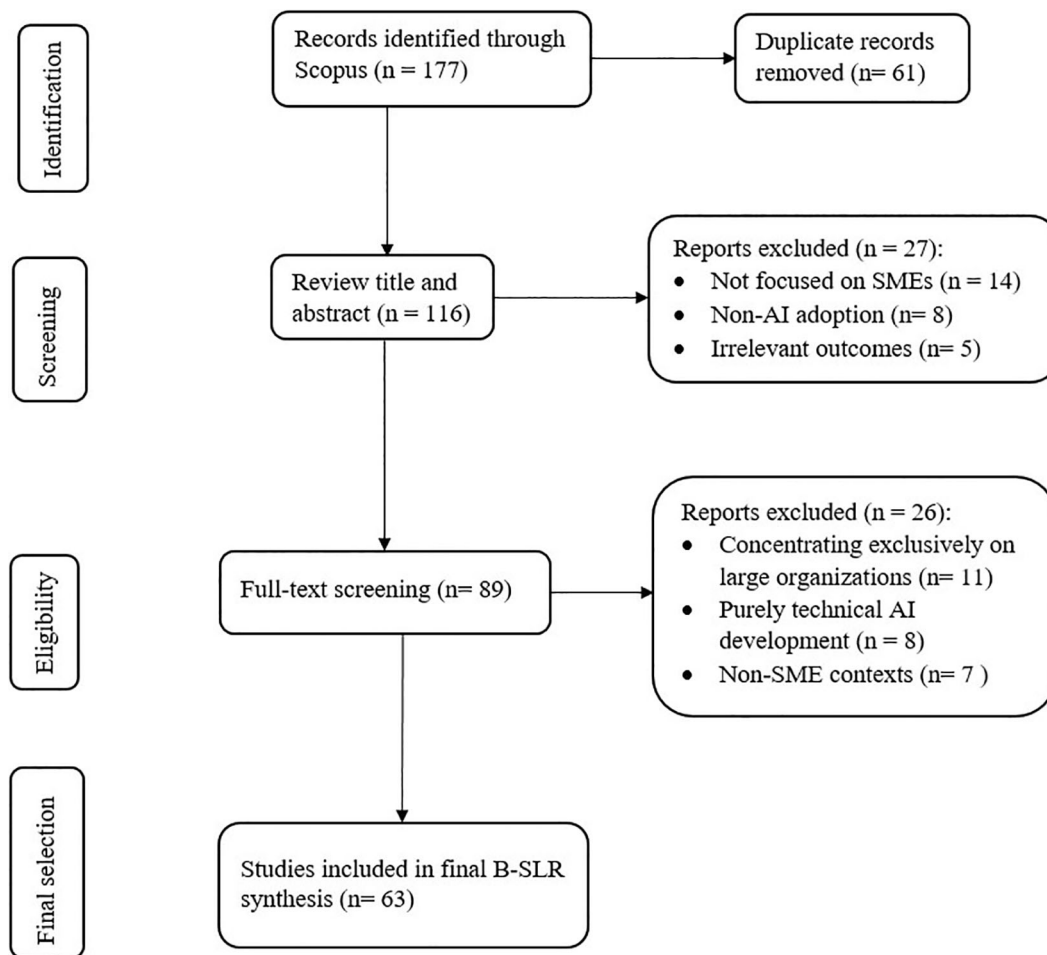


Figure 1. PRISMA framework for this B-SLR, adapted from Page et al. (2021).

thematic analysis allowed for a comprehensive understanding of SMEs' adoption of AI. This dual approach highlighted the field's intellectual structure and context-specific insights, including sectoral differences, geographical gaps, and adoption drivers. The methodology strengthens reliability and practical relevance, providing a solid basis to guide future research, support managerial decisions, and inform policies that promote AI-driven innovation and competitiveness in SMEs.

Results

The bibliometric analysis reveals a clear and accelerating trajectory of scholarly interest in AI adoption in SMEs over the past two decades. Between 2010 and 2015, research on this topic was sparse and largely conceptual, focusing on theoretical models of technology adoption and organizational readiness (Cunningham et al., 2023). Studies during this period often extrapolated findings from larger firms to SMEs, with limited empirical validation. From 2016 to 2020, the number of publications increased moderately, reflecting the early adoption of Industry 4.0 initiatives and the growing recognition of digital transformation as a strategic priority for SMEs (Kee et al., 2025; Pelletier & Cloutier, 2019; Yaqub & Alsabban, 2023). Post-2020, there has been a marked surge in publications, coinciding with widespread AI integration into business operations and heightened attention to innovation, strategic planning, and competitive advantage (Carayannis et al., 2025). This trend highlights both the growing academic interest and the alignment of research with practical business needs in digitally evolving economies.

Bibliographic coupling analysis at the country level reveals that scholarship in the field is largely concentrated in Europe and North America, with notable contributions from the United Kingdom, Germany, the United States, and Canada (Ayinaddis, 2025; Oldemeyer et al., 2025). Patterns of co-authorship suggest the expansion of collaborative research networks; however, the involvement of emerging economies remains limited (see Figures 2 and 3). This regional imbalance underscores a significant knowledge gap regarding SMEs operating within contexts characterized by resource constraints, regulatory diversity, and limited digital infrastructure (Aderibigbe et al., 2023).

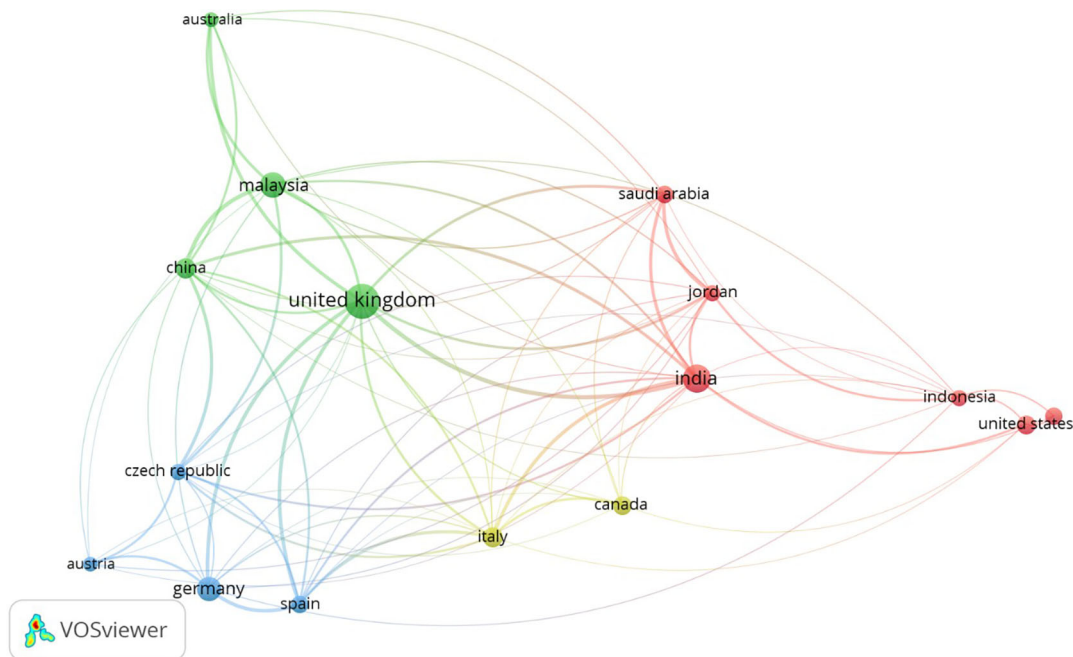


Figure 2. Bibliographic coupling analysis of country-level contributions.

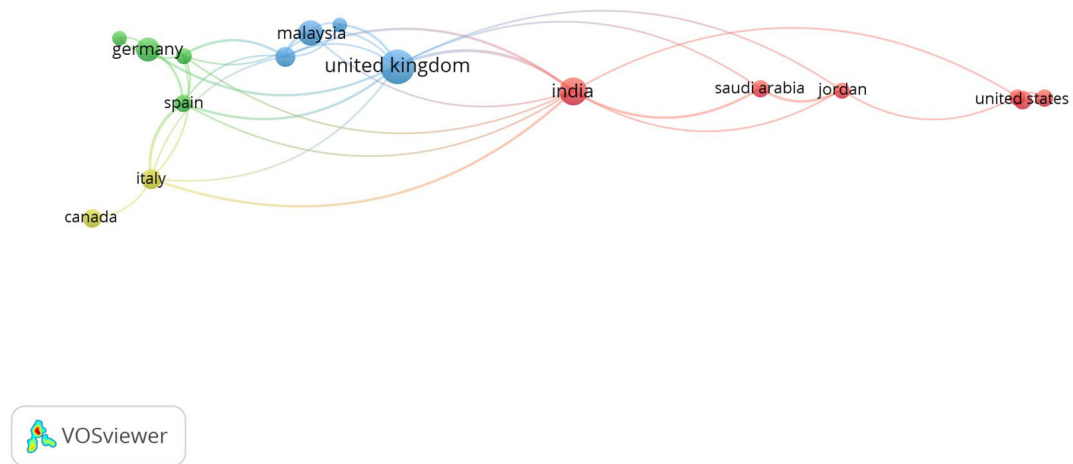


Figure 3. Co-authorship networks across countries.

Keyword co-occurrence and thematic mapping identify three principal clusters (see [Figure 4](#) and [Table 2](#)). The first, AI-driven innovation, highlights the contribution of machine learning, predictive analytics, and process automation to improving operational efficiency, product development, and competitiveness ([Jalil et al., 2025](#); [Wei & Pardo, 2022](#)). The second, entrepreneurial decision-making, emphasizes how AI facilitates opportunity recognition, evidence-based strategic planning, and agile responses to market uncertainty ([Ikbāl, 2025](#); [Vudugula et al., 2023](#)). The third, adoption barriers and enablers, encompasses internal and external factors, such as human capital, technological infrastructure, regulatory conditions, and integration into digital ecosystems ([Prasad Agrawal, 2024](#); [Trevisan et al., 2023](#)). Studies consistently identify financial constraints, limited digital literacy, and weak organizational support as obstacles to AI adoption. At the same time, leadership commitment, inter-organizational partnerships, and targeted training emerge as key enabling mechanisms.

The systematic literature review extends the bibliometric insights by offering empirical evidence on the outcomes of AI adoption. Findings indicate that AI enhances operational efficiency, reduces costs, streamlines supply chains, and improves customer engagement ([Iyelolu et al., 2024](#)). Furthermore, AI catalyzes process and product innovation,

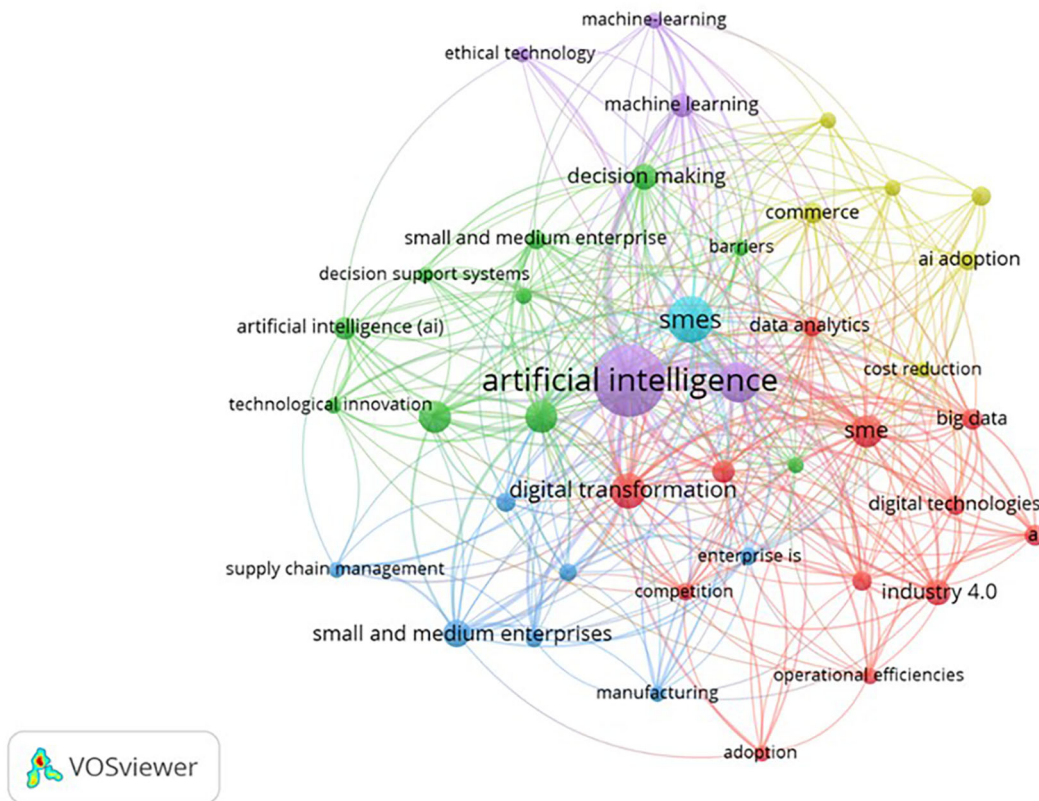


Figure 4. Keyword co-occurrence mapping.

Table 2. Key research clusters and representative keywords.

Cluster	Representative keywords	Focus area
AI-driven Innovation	Machine learning, predictive analytics, automation, process optimization	Operational efficiency, product innovation, and competitiveness
Entrepreneurial Decision-Making	Opportunity recognition, strategic planning, market sensing	Evidence-based managerial decisions, risk management
Adoption Barriers/Enablers	Organizational readiness, human capital, financial resources, and digital ecosystems	Determinants of adoption, enabling strategies

supporting SMEs in developing new offerings, reconfiguring business models, and achieving competitive differentiation (Sjödin et al., 2023; Tetteh et al., 2025). It also enhances entrepreneurial decision-making by leveraging predictive analytics and real-time insights for market sensing, risk assessment, and informed strategic investment. Nonetheless, the impact of AI adoption is highly contingent upon factors such as firm size, industry characteristics, managerial capabilities, and regional context (Horani et al., 2025).

Methodologically, the empirical literature is dominated by case studies and survey-based research, which account for the majority of existing studies. In contrast, longitudinal and experimental approaches remain limited, constraining insights into the long-term effects of AI adoption on innovation and competitiveness (Schwaake et al., 2025). Moreover, the evidence base disproportionately centers on SMEs in Europe and North America, with considerably less attention given to those in Asia, Africa, and Latin America. This geographic imbalance underscores the need for context-sensitive research that addresses the technological, institutional, and resource-related challenges faced by SMEs in emerging economies (Dey et al., 2024; Kumar et al., 2024).

In conclusion, the combined insights from bibliometric and systematic review analyses offer a comprehensive perspective on AI adoption within SMEs. The findings indicate a growing scholarly focus, with innovation, decision-making

Table 3. Thematic clusters, associated sub-domains, core focus areas, and illustrative studies.

Cluster theme	Sub-themes	Key focus	Representative studies
AI-driven Innovation	- Machine Learning - Predictive Analytics - Automation - Process Optimization	Explores how AI technologies enhance innovation through improved processes, automation, and predictive insights.	(Ali Abbasi et al., 2022; Baabdullah et al., 2021; Barata et al., 2023; Belahouaoui & Attak, 2024; Bettoni et al., 2021; Chatterjee et al., 2022; Crockett et al., 2021; Dey et al., 2024; Hui et al., 2024; Le et al., 2020; Mathani et al., 2024; Metin et al., 2024; Musa & Lefkir, 2024; Panigrahi et al., 2023; Parra-Sánchez & Talero-Sarmiento, 2024; Rawashdeh, 2022; Taherizadeh & Beaudry, 2023; Tominc et al., 2024; Wei & Pardo, 2022)
Entrepreneurial Decision-Making	- Opportunity Recognition - Strategic Planning - Market Sensing	Examines how AI supports entrepreneurs in identifying opportunities, making strategic choices, and sensing market trends.	(Alaskari et al., 2021; Alsafi & Fan, 2020; Baabdullah et al., 2021; Baldegger et al., 2020; Ejami, 2024; Enshassi et al., 2024; Etemad, 2023; Hermansyah, 2023; Islam et al., 2024a, b; Kshetri et al., 2025; Magableh et al., 2024; Muminova et al., 2024; Polas et al., 2022; Pozzo et al., 2024; Proenca, 2024; Roux et al., 2023; Saleem et al., 2024; Tawil et al., 2024; Vranda et al., 2021; Zavodna et al., 2024)
Adoption Barriers/Enablers	- Organizational Readiness - Human Capital - Financial Resources - Digital Ecosystems	Investigate internal and external factors that facilitate or hinder the adoption of AI in SMEs, including resources, skills, and ecosystem support.	(Aljarboa, 2024; Almashawreh et al., 2024; Anh & Duc, 2024; Arroyabe et al., 2024; Badghish & Soomro, 2024; Chatterjee et al., 2022; Díaz-Arancibia et al., 2024; Fonseka et al., 2022; Govori & Sejdija, 2023; Gupta, 2024; Gupta & Yang, 2024; Hamdan et al., 2022; Jin & Liu, 2025; Kukanja, 2024; Lada et al., 2023; Madhavan et al., 2022; Muto et al., 2024; Polisetty et al., 2024; Rawindaran et al., 2021; Roux et al., 2023; Saleem et al., 2023)

processes, and adoption drivers identified as key thematic areas. At the same time, notable methodological and regional limitations remain. These outcomes offer valuable guidance for researchers, practitioners, and policymakers seeking to promote AI-enabled innovation, enhance entrepreneurial decision-making, and boost the global competitiveness of SMEs.

Table 3 provides a structured overview of three central cluster themes in the study of AI adoption in SMEs: AI-driven innovation, entrepreneurial decision-making, and adoption barriers/enablers. Each theme is further broken down into sub-themes that capture specific dimensions of AI application and organizational dynamics. For AI-driven innovation, sub-themes such as machine learning, predictive analytics, automation, and process optimization illustrate the technological mechanisms through which SMEs enhance product, service, and operational innovation. Within the entrepreneurial decision-making cluster, opportunity recognition, strategic planning, and market sensing highlight how AI supports informed, agile, and market-oriented managerial decisions. Finally, the adoption barriers/enablers theme emphasizes organizational readiness, human capital, financial resources, and digital ecosystems as key factors that shape the successful implementation of AI solutions. The table synthesizes key focus areas and representative studies, offering scholars and practitioners a concise framework to understand how AI technologies, entrepreneurial processes, and organizational capabilities interact to drive SME performance.

Discussion

The results of this study provide compelling evidence of AI's transformative impact on SMEs, with significant implications for innovation, managerial decision-making, and strategic competitiveness. By integrating insights from a B-SLR, the study identifies key thematic areas, prevailing methodological trends, and existing research gaps, offering a comprehensive understanding of AI adoption in SMEs. The bibliometric analysis indicates a marked increase in scholarly

attention to AI adoption in SMEs since 2020, reflecting global shifts toward digital transformation, Industry 4.0, and the growing recognition of AI as a strategic enabler (Khan et al., 2025; Schwaeke et al., 2025). This trend suggests that SMEs are increasingly positioned at the intersection of technological innovation and entrepreneurship, leveraging AI to enhance operational efficiency and create strategic value. The identified thematic clusters, AI-driven innovation, entrepreneurial decision-making, and adoption barriers/enablers, highlight the multidimensional nature of AI integration within SMEs (Upadhyay et al., 2023; Wang et al., 2022).

AI-driven innovation emerges as a central outcome of adoption. Literature consistently shows that SMEs leveraging AI technologies experience enhanced product and process innovation, faster time-to-market, and improved operational efficiency (Cooper, 2025). These findings align with the dynamic capabilities perspective, which emphasizes the importance of leveraging technology to sense and seize market opportunities, reconfigure resources, and achieve a sustainable competitive advantage (Drydakis, 2022). SMEs that integrate AI can anticipate market shifts, customize their offerings, and optimize resource allocation, thereby positioning themselves as agile and innovative actors within competitive environments (Carayannis et al., 2025). This underscores the strategic potential of AI adoption, enabling SMEs to overcome traditional resource constraints by harnessing intelligent technologies.

The study also highlights the role of AI in supporting entrepreneurial decision-making. Predictive analytics, data-driven insights, and decision-support systems enhance the capacity of SMEs to identify business opportunities, evaluate risks, and make informed strategic choices (Žilka et al., 2024). This indicates that AI functions not only as a technological tool but also as an enabler of managerial cognition and strategic foresight. By facilitating evidence-based decisions, AI adoption allows SMEs to navigate uncertainty, respond effectively to market disruptions, and align entrepreneurial initiatives with emerging business trends (Siddiqui et al., 2024). Consequently, AI adoption is intricately linked to the development of entrepreneurial capabilities and fostering resilience and adaptability in small, resource-constrained organizations.

Despite these benefits, adoption is shaped by a complex interplay of internal and external factors. Organizational readiness, encompassing technological infrastructure, managerial expertise, and digital literacy, is crucial in determining the success of adoption (Adam & Alarifi, 2021; Kumar & Sharma, 2023). SMEs often face barriers such as high implementation costs, a lack of skilled personnel, and limited awareness of AI applications (Al Dhaheri et al., 2024). Externally, regulatory frameworks, industry standards, and access to collaborative networks influence the extent and impact of AI adoption (Arroyabe et al., 2024; Chatterjee, 2020). These findings suggest that adoption is not a linear process, but rather requires tailored strategies that account for the unique contexts of SMEs across different regions and sectors.

From a methodological perspective, the study relies strongly on case studies and surveys, which provide rich qualitative and quantitative insights but limit longitudinal understanding of AI adoption outcomes (Bedué & Fritzsche, 2022; Polisetty et al., 2024; Uren & Edwards, 2023). The scarcity of longitudinal, experimental, and mixed-method studies represents a significant gap, particularly in understanding the sustained impact of AI on innovation performance, financial outcomes, and strategic decision-making. Additionally, the research concentration in developed economies limits generalizability to SMEs in emerging markets, where resource constraints, infrastructure limitations, and contextual challenges may influence adoption trajectories differently (Ashiru et al., 2023; Elbanna et al., 2020). Therefore, targeted empirical studies in underrepresented regions and diverse sectors are crucial for a comprehensive understanding of AI adoption globally in SMEs.

Synthesizing B-SLR findings highlights strategic research opportunities. Future studies could examine sector-specific AI adoption pathways, particularly in service and non-manufacturing SMEs, where digital transformation is increasingly critical. Longitudinal and experimental designs are necessary to assess the sustained impact of AI on innovation, decision-making, and firm performance. Research in emerging economies can shed light on the role of institutional support, funding mechanisms, and local entrepreneurial ecosystems in facilitating the adoption of innovative solutions. Ultimately, integrating multi-level analyses that consider individual, organizational, and environmental determinants can yield nuanced insights, thereby advancing both theory and practice.

AI adoption in SMEs is a multidimensional phenomenon shaped by technological capabilities, managerial competencies, organizational culture, and environmental factors. Strategic AI integration fosters innovation, supports entrepreneurial decision-making, and enhances competitive resilience, while challenges persist regarding resource, knowledge, and institutional support. These insights offer significant implications for scholars, practitioners, and policymakers aiming to transform SMEs into agile, innovative, and competitive entities in rapidly evolving business environments.

Future directions and implications

The findings of this study highlight several critical directions for advancing research on AI adoption in SMEs. First, there is a clear need for longitudinal studies to assess the sustained impact of AI on organizational performance, innovation outcomes, and entrepreneurial decision-making. Much existing research relies on cross-sectional designs, which overlook dynamic adoption trajectories and long-term effects. Longitudinal approaches would generate richer insights into how SMEs evolve with AI integration, shedding light on how technological adoption fosters strategic agility, resilience, and sustained competitiveness (Ayinaddis, 2025).

Second, research should explore sector-specific pathways of AI adoption. While manufacturing SMEs dominate existing studies, service-oriented, retail, and creative industries remain significantly underexplored (Wang & Zhang, 2025). Examining AI integration across diverse sectors will illuminate how technological adaptation, organizational processes, and innovation outputs differ in various contexts, offering nuanced guidance for managers seeking to leverage AI for value creation. This sectoral lens is critical as AI adoption increasingly intersects with digital services, customer experience, and knowledge-intensive business models.

Third, there is a critical need for contextualized research in emerging economies. Most existing studies focus on SMEs in Europe and North America, leaving a substantial knowledge gap regarding adoption barriers, institutional support, and innovation outcomes in Africa, Latin America, and Asia (Balzano et al., 2025; Torres de Oliveira et al., 2022). Comparative studies across regions can reveal how infrastructural limitations, regulatory environments, cultural factors, and entrepreneurial ecosystems shape AI adoption and its outcomes. Such research would broaden the generalizability of the findings and provide actionable insights for policymakers and international development agencies seeking to enhance SME competitiveness in resource-constrained environments.

Future research should adopt a multi-level approach to better understand the adoption of AI by SMEs. This involves examining factors at the individual level (such as managerial skills and employee AI literacy), the organizational level (including culture and resources), and the environmental level (including knowledge networks and regulations). Considering these factors together would provide a more comprehensive picture of how AI adoption creates value (Dwivedi et al., 2021). Using experimental or mixed-method approaches would also improve causal insights, helping researchers link AI adoption more directly to innovation, decision-making, and firm performance.

Theoretically, this study reinforces and extends the dynamic capabilities framework by demonstrating that AI serves as a strategic resource, enabling SMEs to sense opportunities, seize market advantages, and reconfigure internal capabilities for innovation and competitiveness (Teece, 2018). The findings highlight the mediating role of organizational readiness, technological infrastructure, and external support mechanisms. This demonstrates that successful adoption is not purely a function of technology but emerges from the interplay of human, organizational, and environmental factors. These insights advance technology adoption, innovation management, and entrepreneurship theory by highlighting the complex interdependencies that shape the adoption of AI in SMEs.

Practically, this study provides guidance to managers on how to enhance the effectiveness of AI adoption. Aligning AI with organizational goals is crucial to unlock its full potential for innovation, improved decision-making, and a competitive advantage, beyond simply enhancing efficiency. Building capacity through employee training, enhancing managerial AI literacy, and hiring skilled personnel are key steps for successful use. SMEs should also evaluate their technological infrastructure, digital maturity, and internal processes to ensure readiness. Additionally, collaborating with industry partners and leveraging external knowledge networks can help overcome adoption barriers and enhance innovation outcomes.

Policymakers also play a pivotal role in enabling AI adoption. Tailored policies addressing SMEs' financial constraints, limited technical expertise, and digital infrastructure gaps can facilitate adoption and maximize their benefits. Initiatives such as tax incentives, subsidized training programs, and regional innovation hubs can provide both the resources and guidance SMEs need. In parallel, clear regulatory frameworks for data privacy, AI ethics, and compliance standards can help reduce uncertainty, fostering the responsible and sustainable integration of AI. By addressing both organizational and institutional barriers, such policies can accelerate SMEs' transformation into agile, innovative, and competitive enterprises.

In summary, future research should use longitudinal, sector-specific, and context-sensitive approaches, combining multi-level and experimental designs to capture both the complexity and long-term effects of AI adoption in SMEs. Theoretically, studies should focus on how technology, managerial capabilities, and environmental factors interact with each other. Practically, attention should be given to strategic integration, capacity building, and supportive ecosystems.

These directions can help SMEs, scholars, and policymakers harness AI for sustainable innovation, agility, and competitiveness in a digital economy. Policymakers and government agencies also play a crucial role in promoting the adoption of AI. Tailored policies are necessary to address the challenges faced by SMEs, including financial constraints, limited expertise, and inadequate digital infrastructure. Possible measures include tax incentives for AI investment, subsidized training, and the establishment of regional innovation hubs. Clear regulations on data privacy, AI ethics, and compliance can further reduce uncertainty and support the responsible adoption of AI.

Conclusion

This study provides a comprehensive and evidence-based understanding of AI adoption in SMEs by integrating bibliometric analysis with a SLR. The bibliometric analysis quantitatively maps the growth of research, influential institutions, thematic clusters, and intellectual networks, revealing a rapidly expanding field concentrated primarily in developed economies. Complementing this, the SLR synthesizes empirical evidence on AI-driven innovation, entrepreneurial decision-making, and adoption barriers and enablers, collectively offering a holistic view of the current state of knowledge while identifying critical gaps that warrant scholarly attention.

The findings highlight that AI adoption serves as a strategic enabler for SMEs, strengthening innovation capabilities, supporting evidence-based decision-making, and enhancing operational efficiency. When effectively leveraged, AI enables SMEs to overcome resource constraints, respond proactively to changing market conditions, and establish a sustainable competitive advantage. However, successful adoption requires organizational readiness, managerial expertise, a robust technological infrastructure, and a supportive external environment. Methodologically, the dominance of case studies and surveys underscores the need for more longitudinal, experimental, and mixed-method research to capture the long-term and context-specific impacts of AI adoption.

Geographical disparities in the literature point to an urgent need for research in emerging economies, where SMEs face distinct challenges and opportunities. Policymakers can accelerate adoption by providing targeted support programs, investing in infrastructure development, and offering digital skills training. At the same time, SME managers can enhance organizational readiness, workforce capabilities, and strategic integration of AI into decision-making processes. For scholars, this study identifies important future research directions, including the exploration of sector-specific adoption patterns, examination of long-term performance outcomes, and investigation of how technological, organizational, and environmental factors interact.

In conclusion, this study makes significant contributions to both theory and practice by offering a holistic understanding of AI adoption in SMEs. It clarifies the intellectual structure, thematic priorities, methodological trends, and research gaps in the field while providing actionable insights for managers and policymakers. By combining bibliometric and systematic review approaches, this study not only maps the current body of knowledge but also provides a foundation for future research and evidence-based interventions to promote innovation, competitiveness, and sustainable growth in SMEs worldwide.

Availability of data and materials

The datasets used in this study are publicly accessible through the Zenodo repository under the title Data Repository Scopus 63, with the working DOI <https://doi.org/10.5281/zenodo.17375011> (Yesuf, 2025).

Data are available under the terms of the [Creative Commons Attribution 4.0 International license](#) (CC-BY 4.0).

References

-
- OECD: *Financing SMEs and Entrepreneurs 2024: An OECD Scoreboard*. Paris: OECD Publishing; 2024.
[Publisher Full Text](#)
- Adam NA, Alarifi G: **Innovation practices for survival of small and medium enterprises (SMEs) in the COVID-19 times: the role of external support**. *J. Innov. Entrep.* 2021; **10**(1): 15.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
- Adekunle BI, Chukwuma-Eke EC, Balogun ED, *et al.*: **Machine learning for automation: Developing data-driven solutions for process optimization and accuracy improvement**. *Mach. Learn.* 2021; **3**(1): 800–808.
[Publisher Full Text](#)
- Aderibigbe AO, Ohenhen PE, Nwaobia NK, *et al.*: **Artificial intelligence in developing countries: Bridging the gap between potential and implementation**. *Computer Science & IT Research Journal.* 2023; **4**(3): 185–199.
[Publisher Full Text](#)
- Al Dhaheri MH, Ahmad SZ, Papastathopoulos A: **Do environmental turbulence, dynamic capabilities, and artificial intelligence force SMEs to be innovative?** *J. Innov. Knowl.* 2024; **9**(3): 100528.
[Publisher Full Text](#)
- Alaskari O, Pinedo-Cuenca R, Ahmad MM: **Framework for implementation of enterprise resource planning (ERP) systems in small and medium enterprises (SMEs): A case study**. *Procedia Manufacturing.* 2021; **55**: 424–430.
[Publisher Full Text](#)
- Albats E, Podmetina D, Vanhaverbeke W: **Open innovation in SMEs: A process view towards business model innovation**. *J. Small Bus. Manag.*

2023; **61**(6): 2519–2560.

[Publisher Full Text](#)

Ali Abbasi G, Abdul Rahim NF, Wu H, *et al.*: **Determinants of SME's social media marketing adoption: competitive industry as a moderator.** *SAGE Open*. 2022; **12**(1): 21582440211067220.

[Publisher Full Text](#)

Aljarboa S: **Factors influencing the adoption of artificial intelligence in e-commerce by small and medium-sized enterprises.** *IJIM Data Insights*. 2024; **4**(2): 100285.

Almashawreh R, Talukder M, Charath SK, *et al.*: **AI adoption in Jordanian SMEs: The influence of technological and organizational orientations.** *Glob. Bus. Rev.* 2024; 09721509241250273.

[Publisher Full Text](#)

Alsafi T, Fan I-S: **Cloud computing adoption barriers faced by Saudi manufacturing SMEs.** *2020 15th Iberian Conference on Information Systems and Technologies (CISTI)*. 2020; 1–6.

Anh DN, Duc PM: **Social responsibility of small and medium enterprises in Vietnam through digital transformation and application of artificial intelligence.** *LatIA*. 2024; **2**: 99.

[Publisher Full Text](#)

Aria M, Cuccurullo C: **bibliometrix: An R-tool for comprehensive science mapping analysis.** *J. Informet.* 2017; **11**(4): 959–975.

[Publisher Full Text](#)

Arroyabe MF, Arranz CFA, De Arroyabe IF, *et al.*: **Analyzing AI adoption in European SMEs: A study of digital capabilities, innovation, and external environment.** *Technol. Soc.* 2024; **79**: 102733.

[Publisher Full Text](#)

Ashiru F, Nakpodia F, You JJ: **Adapting emerging digital communication technologies for resilience: evidence from Nigerian SMEs.** *Ann. Oper. Res.* 2023; **327**(2): 795–823.

[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)

Ayinaddis SG: **Artificial intelligence adoption dynamics and knowledge in SMEs and large firms: A systematic review and bibliometric analysis.** *J. Innov. Knowl.* 2025; **10**(3): 100682.

[Publisher Full Text](#)

Baabdullah AM, Alalwan AA, Slade EL, *et al.*: **SMEs and artificial intelligence (AI): Antecedents and consequences of AI-based B2B practices.** *Ind. Mark. Manag.* 2021; **98**: 255–270.

[Publisher Full Text](#)

Badghish S, Soomro YA: **Artificial intelligence adoption by SMEs to achieve sustainable business performance: application of technology–organization–environment framework.** *Sustainability*. 2024; **16**(5): 1864.

[Publisher Full Text](#)

Baldegger R, Caon M, Sadiku K: **Correlation between entrepreneurial orientation and implementation of AI in human resources management (HRM).** *Technol. Innov. Manag. Rev.* 2020.

Balzano M, Marzi G, Turzo T: **SMEs and institutional theory: major inroads and opportunities ahead.** *Manag. Decis.* 2025; **63**(13): 1–27.

[Publisher Full Text](#)

Barata SPFG, Ferreira FAF, Carayannis EG, *et al.*: **Determinants of E-commerce, artificial intelligence, and agile methods in small and medium-sized enterprises.** *IEEE Trans. Eng. Manag.* 2023; **71**: 6903–6917.

[Publisher Full Text](#)

Bedué P, Fritzsche A: **Can we trust AI? An empirical investigation of trust requirements and guide to successful AI adoption.** *J. Enterp. Inf. Manag.* 2022; **35**(2): 530–549.

[Publisher Full Text](#)

Belahouaoui R, Attak EH: **Digital taxation, artificial intelligence and Tax Administration 3.0: Improving tax compliance behavior—a systematic literature review using textometry (2016–2023).** *Accounting Research Journal*. 2024; **37**(2): 172–191.

[Publisher Full Text](#)

Bettoni A, Matteri D, Montini E, *et al.*: **An AI adoption model for SMEs: A conceptual framework.** *IFAC-PapersOnLine*. 2021; **54**(1): 702–708.

[Publisher Full Text](#)

Carayannis EG, Dumitrescu R, Falkowski T, *et al.*: **Enhancing SME resilience through artificial intelligence and strategic foresight: A framework for sustainable competitiveness.** *Technol. Soc.* 2025; **81**: 102835.

[Publisher Full Text](#)

Castillo-Vergara M, Duarte Valdivia D, Muñoz-Cisterna V, *et al.*: **Digital capabilities of SMEs: driving the Industry 4.0 revolution and measuring its innovative effects.** *Academia Revista Latinoamericana de Administración*. 2025; **38**(1): 74–105.

[Publisher Full Text](#)

Chatterjee S: **AI strategy of India: policy framework, adoption challenges and actions for government.** *Transforming Government: People, Process and Policy*. 2020; **14**(5): 757–775.

[Publisher Full Text](#)

Chatterjee S, Chaudhuri R, Vrontis D, *et al.*: **Digital transformation and entrepreneurship process in SMEs of India: a moderating role of**

adoption of AI-CRM capability and strategic planning. *J. Strateg. Manag.* 2022; **15**(3): 416–433.

[Publisher Full Text](#)

Cooper RG: **SMEs' use of AI for new product development: Adoption rates by application and readiness-to-adopt.** *Ind. Mark. Manag.* 2025; **126**: 159–167.

[Publisher Full Text](#)

Crockett K, Colyer E, Gerber L, *et al.*: **Building trustworthy AI solutions: A case for practical solutions for small businesses.** *IEEE Transactions on Artificial Intelligence*. 2021; **4**(4): 778–791.

[Publisher Full Text](#)

Cunningham JA, Damij N, Modic D, *et al.*: **MSME technology adoption, entrepreneurial mindset and value creation: a configurational approach.** *J. Technol. Transf.* 2023; **48**(5): 1574–1598.

[Publisher Full Text](#)

Dey PK, Chowdhury S, Abadie A, *et al.*: **Artificial intelligence-driven supply chain resilience in Vietnamese manufacturing small and medium-sized enterprises.** *Int. J. Prod. Res.* 2024; **62**(15): 5417–5456.

[Publisher Full Text](#)

Díaz-Arancibia J, Hochstetter-Díez J, Bustamante-Mora A, *et al.*: **Navigating digital transformation and technology adoption: A literature review from small and medium-sized enterprises in developing countries.** *Sustainability*. 2024; **16**(14): 5946.

[Publisher Full Text](#)

Donthu N, Kumar S, Mukherjee D, *et al.*: **How to conduct a bibliometric analysis: An overview and guidelines.** *J. Bus. Res.* 2021; **133**: 285–296.

[Publisher Full Text](#)

Drydakis N: **Artificial Intelligence and reduced SMEs' business risks. A dynamic capabilities analysis during the COVID-19 pandemic.** *Inf. Syst. Front.* 2022; **24**(4): 1223–1247.

[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)

Dwivedi YK, Hughes L, Ismagilova E, *et al.*: **Artificial Intelligence (AI): Multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy.** *Int. J. Inf. Manag.* 2021; **57**: 101994.

[Publisher Full Text](#)

Ejjami R: **Adopting artificial intelligence and big data tools across industry sectors in Morocco: an integrative literature review.** *Int. J. Environ. Workplace Employ.* 2024; **8**(2): 171–198.

[Publisher Full Text](#)

Elbanna S, Hsieh L, Child J: **Contextualizing internationalization decision-making research in SMEs: Towards an integration of existing studies.** *Eur. Manag. Rev.* 2020; **17**(2): 573–591.

[Publisher Full Text](#)

Enshassi M, Nathan RJ, Soekmawati S, *et al.*: **Potentials of artificial intelligence in digital marketing and financial technology for small and medium enterprises.** *IAES International Journal of Artificial Intelligence (IJ-AI)*. 2024; **13**(1): 639.

[Publisher Full Text](#)

Etamad H: **The increasing prevalence of multi-sided online platforms and their influence on international entrepreneurship: The rapid transformation of entrepreneurial digital ecosystems.** *J. Int. Entrep.* 2023; **21**(1): 1–30.

[Publisher Full Text](#)

Feher K: *Generative AI, Media, and Society*. Taylor & Francis; 2025.

Ferreira J, Cardim S, Coelho A: **Dynamic capabilities and mediating effects of innovation on the competitive advantage and firm's performance: The moderating role of organizational learning capability.** *J. Knowl. Econ.* 2021; **12**(2): 620–644.

[Publisher Full Text](#)

Fonseka K, Jaharadak AA, Raman M: **Impact of E-commerce adoption on business performance of SMEs in Sri Lanka; moderating role of artificial intelligence.** *Int. J. Soc. Econ.* 2022; **49**(10): 1518–1531.

[Publisher Full Text](#)

Ghobakhloo M, Iranmanesh M, Tseng M-L, *et al.*: **Behind the definition of Industry 5.0: a systematic review of technologies, principles, components, and values.** *J. Ind. Prod. Eng.* 2023; **40**(6): 432–447.

[Publisher Full Text](#)

Giuglioli G, Pellegrini MM: **Artificial intelligence as an enabler for entrepreneurs: a systematic literature review and an agenda for future research.** *Int. J. Entrep. Behav. Res.* 2023; **29**(4): 816–837.

[Publisher Full Text](#)

Govori A, Sejdija Q: **Future prospects and challenges of integrating artificial intelligence within the business practices of small and medium enterprises.** *Governance & Regulation*. 2023; **12**: 176–183.

[Publisher Full Text](#)

Gupta V: **An empirical evaluation of a generative artificial intelligence technology adoption model from entrepreneurs' perspectives.** *Systems*. 2024; **12**(3): 103.

[Publisher Full Text](#)

Gupta V, Yang H: **Study protocol for factors influencing the adoption of ChatGPT technology by startups: Perceptions and attitudes of**

entrepreneurs. *Plos One*. 2024; **19**(2): e0298427.

[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)

Hamdan IKA, Sumarliah E, Fauziyah F: **A machine learning method to predict the technology adoption of blockchain in Palestinian firms.** *Int. J. Emerg. Mark.* 2022; **17**(4): 1008–1029.

[Publisher Full Text](#)

Hassan W, Duarte AE: **Bibliometric analysis: a few suggestions.** *Curr. Probl. Cardiol.* 2024; **49**(8): 102640.

[PubMed Abstract](#) | [Publisher Full Text](#)

Hermansyah Y: **Assessing the impact of communicative artificial intelligence based accounting information systems on small and medium enterprises.** *Journal of Wireless Mobile Networks, Ubiquitous Computing, and Dependable Applications.* 2023; **14**(3): 230–239.

[Publisher Full Text](#)

Horani OM, Al-Adwan AS, Yaseen H, et al.: **The critical determinants impacting artificial intelligence adoption at the organizational level.** *Inf. Dev.* 2025; **41**(3): 1055–1079.

[Publisher Full Text](#)

Hui L, Luo Z, Liu K: **Impact of pollution prevention practices and green environmental practices on sustainable performance: Empirical evidence from Chinese SMEs.** *Environ. Res.* 2024; **255**: 118991.

[PubMed Abstract](#) | [Publisher Full Text](#)

Ikbal MZ: **A META-ANALYSIS OF AI-DRIVEN BUSINESS ANALYTICS: ENHANCING STRATEGIC DECISION-MAKING IN SMES.** *Review of Applied Science and Technology.* 2025; **04**(02): 33–59.

[Publisher Full Text](#)

Islam MA, Fakir SI, Masud SB, et al.: **Artificial intelligence in digital marketing automation: Enhancing personalization, predictive analytics, and ethical integration.** *Edelweiss Appl. Sci. Technol.* 2024a; **8**(6): 6498–6516.

[Publisher Full Text](#)

Islam MT, Kumar J, Konar R: **Small Steps, Big Impact: Conceptualising the Adoption of Social Media Marketing in Bangladesh's SMEs.** *AI Impacts in Digital Consumer Behavior.* IGI Global Scientific Publishing; 2024b; pp. 217–240.

Iyelolu TV, Agu EE, Idemudia C, et al.: **Driving SME innovation with AI solutions: overcoming adoption barriers and future growth opportunities.** *International Journal of Science and Technology Research Archive.* 2024; **7**(1): 036–054.

[Publisher Full Text](#)

Jalil MF, Lynch P, Marikan DABA, et al.: **The influential role of artificial intelligence (AI) adoption in digital value creation for small and medium enterprises (SMEs): does technological orientation mediate this relationship?** *AI & SOCIETY.* 2025; **40**(3): 1875–1896.

[Publisher Full Text](#)

Jin L, Liu M: **Unlocking financial opportunities: The substantial alleviation of financing constraints on small and micro enterprises through digital inclusive finance.** *J. Knowl. Econ.* 2025; **16**(1): 2283–2309.

[Publisher Full Text](#)

Kee DMH, Cordova M, Khin S: **The key enablers of SMEs readiness in Industry 4.0: a case of Malaysia.** *Int. J. Emerg. Mark.* 2025; **20**(3): 1042–1062.

[Publisher Full Text](#)

Khan SAR, Sheikh AA, Shamsi IR, et al.: **The implications of artificial intelligence for small and medium-sized enterprises' sustainable development in the areas of blockchain technology, supply chain resilience, and closed-loop supply chains.** *Sustainability.* 2025; **17**(1): 334.

[Publisher Full Text](#)

Kraus S, Bouncken RB, Yela Aránega A: **The burgeoning role of literature review articles in management research: an introduction and outlook.** *Rev. Manag. Sci.* 2024; **18**(2): 299–314.

[Publisher Full Text](#)

Kshetri N, Rojas-Torres D, Hanafi MM, et al.: **Harnessing generative artificial intelligence: a game-changer for small and medium enterprises.** *IT Professional.* 2025; **26**(6): 84–89.

[Publisher Full Text](#)

Kukanja M: **Examining the Impact of Entrepreneurial Orientation, Self-Efficacy, and Perceived Business Performance on Managers' Attitudes Towards AI and Its Adoption in Hospitality SMEs.** *Systems.* 2024; **12**(12): 526.

[Publisher Full Text](#)

Kulkov I, Kulkova J, Rohrbeck R, et al.: **Artificial intelligence-driven sustainable development: Examining organizational, technical, and processing approaches to achieving global goals.** *Sustain. Dev.* 2024; **32**(3): 2253–2267.

[Publisher Full Text](#)

Kumar L, Sharma RK: **Exploring research issues and potential future directions in industry 4.0 adoption in SMEs: a comprehensive analysis and integration.** *FIIB Business Review.* 2023; 23197145231203510.

Kumar M, Raut RD, Mangla SK, et al.: **The adoption of artificial intelligence powered workforce management for effective revenue growth of micro, small, and medium scale enterprises (MSMEs).** *Prod. Plan. Control.* 2024; **35**(13): 1639–1655.

[Publisher Full Text](#)

Lada S, Chekima B, Karim MRA, et al.: **Determining factors related to artificial intelligence (AI) adoption among Malaysia's small and medium-sized businesses.** *J. Open Innov. Technol. Mark. Complex.* 2023; **9**(4): 100144.

[Publisher Full Text](#)

Le CH, Le DT, Arey D, et al.: **Challenges and conceptual framework to develop heavy-load manipulators for smart factories.** *International Journal of Mechatronics and Applied Mechanics.* 2020; **8**(2): 209–216.

Madhavan M, Wangtueai S, Sharafuddin MA, et al.: **The precipitative effects of pandemic on open innovation of SMEs: a scientometrics and systematic review of Industry 4.0 and Industry 5.0.** *J. Open Innov. Technol. Mark. Complex.* 2022; **8**(3): 152.

[Publisher Full Text](#)

Magableh IK, Mahrouq MH, Ta'Amna MA, et al.: **The role of marketing artificial intelligence in enhancing sustainable financial performance of medium-sized enterprises through customer engagement and data-driven decision-making.** *Sustainability.* 2024; **16**(24): 11279.

[Publisher Full Text](#)

Marzi G, Balzano M, Caputo A, et al.: **Guidelines for Bibliometric-Systematic Literature Reviews: 10 steps to combine analysis, synthesis and theory development.** *Int. J. Manag. Rev.* 2025; **27**(1): 81–103.

[Publisher Full Text](#)

Mathani B, Ajrash HS, Dalaeen AB, et al.: **Identifying variables influencing the adoption of artificial intelligence big data analytics among SMEs in Jordan.** *International Journal of Data and Network Science.* 2024; **8**(4): 2615–2626.

[Publisher Full Text](#)

Metin B, Özhan FG, Wynn M: **Digitalisation and Cybersecurity: Towards an Operational Framework.** *Electronics.* 2024; **13**(21): 4226.

[Publisher Full Text](#)

Muminova E, Ashurov M, Akhunova S, et al.: **AI in small and medium enterprises: Assessing the barriers, benefits, and socioeconomic impacts.** *2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS).* 2024; **1**: 1–6.

Musa AMH, Lefkir H: **The role of artificial intelligence in achieving auditing quality for small and medium enterprises in the Kingdom of Saudi Arabia.** *International Journal of Data and Network Science.* 2024; **8**(2): 835–844.

[Publisher Full Text](#)

Muto V, Luongo S, Percuoco M, et al.: **Artificial intelligence and the great reset: Impacts and perspectives for Italian SMEs business model innovation.** *Systems.* 2024; **12**(9): 330.

[Publisher Full Text](#)

Nazir MA, Rizwan H, Zhu X: **A thematic analysis of factors influencing small and medium enterprise adoption of social media marketing: a TOE framework perspective.** *Qual. Mark. Res. Int. J.* 2025; **28**(1): 178–204.

[Publisher Full Text](#)

Ndubisi NO, Zhai XA, Lai K: **Small and medium manufacturing enterprises and Asia's sustainable economic development.** *Int. J. Prod. Econ.* 2021; **233**: 107971.

[Publisher Full Text](#)

Oldemeyer L, Jede A, Teuteberg F: **Investigation of artificial intelligence in SMEs: a systematic review of the state of the art and the main implementation challenges.** *Manag. Rev. Q.* 2025; **75**(2): 1185–1227.

[Publisher Full Text](#)

Omwole BM, Olufemi-Philips AQ, Ofadile OC, et al.: **Barriers and drivers of digital transformation in SMEs: A conceptual analysis.** *International Journal of Frontline Research in Multidisciplinary Studies.* 2024; **5**(2): 19–36.

Omrani N, Rejeb N, Maalaoui A, et al.: **Drivers of digital transformation in SMEs.** *IEEE Trans. Eng. Manag.* 2022; **71**: 5030–5043.

[Publisher Full Text](#)

Page MJ, McKenzie JE, Bossuyt PM, et al.: **The PRISMA 2020 statement: an updated guideline for reporting systematic reviews.** *BMJ.* 2021; **372**.

Panigrahi RR, Shrivastava AK, Qureshi KM, et al.: **AI Chatbot adoption in SMEs for sustainable manufacturing supply chain performance: a mediational research in an emerging country.** *Sustainability.* 2023; **15**(18): 13743.

[Publisher Full Text](#)

Parra-Sánchez DT, Talero-Sarmiento LH: **Digital transformation in small and medium enterprises: a scientometric analysis.** *Digital Transformation and Society.* 2024; **3**(3): 257–276.

[Publisher Full Text](#)

Pelletier C, Cloutier LM: **Conceptualising digital transformation in SMEs: an ecosystemic perspective.** *J. Small Bus. Enterp. Dev.* 2019; **26**(6/7): 855–876.

[Publisher Full Text](#)

Polas MRH, Jahanshahi AA, Kabir AI, et al.: **Artificial intelligence, blockchain technology, and risk-taking behavior in the 4.0 IR Metaverse Era: evidence from Bangladesh-based SMEs.** *J. Open Innov. Technol. Mark. Complex.* 2022; **8**(3): 168.

[Publisher Full Text](#)

- Polisetty A, Chakraborty D, Sowmya G, *et al.*: **What determines AI adoption in companies? Mixed-method evidence.** *J. Comput. Inf. Syst.* 2024; **64**(3): 370–387.
[Publisher Full Text](#)
- Pozzo DN, Beleño CAG, Correa KR, *et al.*: **Managers' attitudes and behavioral intentions towards using artificial intelligence for organizational decision-making: A study with Colombian SMEs.** *Procedia Computer Science.* 2024; **238**: 956–961.
[Publisher Full Text](#)
- Prasad Agrawal K: **Towards adoption of generative AI in organizational settings.** *J. Comput. Inf. Syst.* 2024; **64**(5): 636–651.
[Publisher Full Text](#)
- Proenca JC: **Business innovation self-assessment with artificial intelligence support for small and medium-sized enterprises.** *Бизнес Управление.* 2024; **4**: 5–17.
- Rawashdeh A: **Determinants of artificial intelligence adoption in SMEs: The mediating role of accounting automation.** 2022.
- Rawindaran N, Jayal A, Prakash E: **Machine learning cybersecurity adoption in small and medium enterprises in developed countries.** *Computers.* 2021; **10**(11): 150.
[Publisher Full Text](#)
- Roux M, Chowdhury S, Kumar Dey P, *et al.*: **Small and medium-sized enterprises as technology innovation intermediaries in sustainable business ecosystem: interplay between AI adoption, low carbon management and resilience.** *Ann. Oper. Res.* 2023; 1–50.
[Publisher Full Text](#)
- Saleem I, Al-Breiki NSS, Asad M: **The nexus of artificial intelligence, frugal innovation and business model innovation to nurture internationalization: A survey of SME's readiness.** *J. Open Innov.: Technol. Mark. Complex.* 2024; **10**(3): 100326.
[Publisher Full Text](#)
- Saleem I, Hoque SMS, Tashfeen R, *et al.*: **The interplay of AI adoption, IoT edge, and adaptive resilience to explain digital innovation: evidence from German family-owned SMEs.** *J. Theor. Appl. Electron. Commer. Res.* 2023; **18**(3): 1419–1430.
[Publisher Full Text](#)
- Sánchez E, Calderón R, Herrera F: **Artificial Intelligence Adoption in SMEs: Survey Based on TOE-DOI Framework, Primary Methodology and Challenges.** *Appl. Sci.* 2025; **15**(12): 6465.
[Publisher Full Text](#)
- Schneegans S, Lewis J, Straza T: *The race against time for smarter development.* Paris: UNESCO; 2021.
- Schwaeke J, Peters A, Kanbach DK, *et al.*: **The new normal: The status quo of AI adoption in SMEs.** *J. Small Bus. Manag.* 2025; **63**(3): 1297–1331.
[Publisher Full Text](#)
- Segarra-Blasco A, Tomàs-Porres J, Teruel M: **AI, robots and innovation in European SMEs.** *Small Bus. Econ.* 2025; **65**: 719–745.
[Publisher Full Text](#)
- Siddiqui D, Mumtaz U, Ahmad N: **Artificial intelligence in entrepreneurship: A bibliometric analysis of the literature.** *J. Glob. Entrep. Res.* 2024; **14**(1): 13.
[Publisher Full Text](#)
- Sjodin D, Parida V, Kohtamäki M: **Artificial intelligence enabling circular business model innovation in digital servitization: Conceptualizing dynamic capabilities, AI capacities, business models and effects.** *Technol. Forecast. Soc. Chang.* 2023; **197**: 122903.
[Publisher Full Text](#)
- Taherizadeh A, Beaudry C: **An emergent grounded theory of AI-driven digital transformation: Canadian SMEs' perspectives.** *Ind. Innov.* 2023; **30**(9): 1244–1273.
[Publisher Full Text](#)
- Tawil A-RH, Mohamed M, Schmoor X, *et al.*: **Trends and challenges towards effective data-driven decision making in UK Small and Medium-sized Enterprises: Case studies and lessons learnt from the analysis of 85 Small and Medium-sized Enterprises.** *Big Data and Cognitive Computing.* 2024; **8**(7): 79.
[Publisher Full Text](#)
- Teece DJ: **Business models and dynamic capabilities.** *Long Range Plan.* 2018; **51**(1): 40–49.
[Publisher Full Text](#)
- Tetteh FK, Gyamerah KK, Nyamekye B, *et al.*: **Digital transformation and business model innovation: the relevance of strategic orientations under varying conditions of competitive intensity.** *J. Manuf. Technol. Manag.* 2025; **36**(3): 621–650.
[Publisher Full Text](#)
- Tominc P, Oreški D, Čančer V, *et al.*: **Statistically Significant Differences in AI Support Levels for Project Management between SMEs and Large Enterprises.** *AI.* 2024; **5**(1): 136–157.
[Publisher Full Text](#)
- Torres de Oliveira R, Gentile-Lüdecke S, Figueira S: **Barriers to innovation and innovation performance: the mediating role of external knowledge search in emerging economies.** *Small Bus. Econ.* 2022; **58**(4): 1953–1974.
[Publisher Full Text](#)
- Trevisan AH, Lobo A, Guzzo D, *et al.*: **Barriers to employing digital technologies for a circular economy: A multi-level perspective.** *J. Environ. Manag.* 2023; **332**: 117437.
[PubMed Abstract](#) | [Publisher Full Text](#)
- Upadhyay N, Upadhyay S, Al-Debei MM, *et al.*: **The influence of digital entrepreneurship and entrepreneurial orientation on intention of family businesses to adopt artificial intelligence: examining the mediating role of business innovativeness.** *Int. J. Entrep. Behav. Res.* 2023; **29**(1): 80–115.
[Publisher Full Text](#)
- Uren V, Edwards JS: **Technology readiness and the organizational journey towards AI adoption: An empirical study.** *Int. J. Inf. Manag.* 2023; **68**: 102588.
[Publisher Full Text](#)
- Vranda J, Tavishi T, Narayanan GB: **Unlocking technology adoption for a robust food supply chain: Evidence from Indian food processing sector.** *Экономический Журнал Высшей Школы Экономики.* 2021; **25**(1): 147–164.
- Vudugula S, Chebrolu SK, Bhuiyan M, *et al.*: **Integrating artificial intelligence in strategic business decision-making: A systematic review of predictive models.** *International Journal of Scientific Interdisciplinary Research.* 2023; **04**(1): 01–26.
[Publisher Full Text](#)
- Wang J, Lu Y, Fan S, *et al.*: **How to survive in the age of artificial intelligence? Exploring the intelligent transformations of SMEs in central China.** *Int. J. Emerg. Mark.* 2022; **17**(4): 1143–1162.
[Publisher Full Text](#)
- Wang K, Sun K, Li Y, *et al.*: **Can digital technology break the financing dilemma of innovative SMEs? Based on the perspective of enterprise innovation.** *Technol. Forecast. Soc. Chang.* 2025; **214**: 124030.
[Publisher Full Text](#)
- Wang S, Zhang H: **Promoting sustainable development goals through generative artificial intelligence in the digital supply chain: Insights from Chinese tourism SMEs.** *Sustain. Dev.* 2025; **33**(1): 1231–1248.
[Publisher Full Text](#)
- Wei R, Pardo C: **Artificial intelligence and SMEs: How can B2B SMEs leverage AI platforms to integrate AI technologies?** *Ind. Mark. Manag.* 2022; **107**: 466–483.
[Publisher Full Text](#)
- World Bank Group: *Global Economic Prospects, January 2022.* World Bank Publications; 2022.
- Yaqub MZ, Alsabban A: **Industry-4.0-enabled digital transformation: Prospects, instruments, challenges, and implications for business strategies.** *Sustainability.* 2023; **15**(11): 8553.
[Publisher Full Text](#)
- Yesuf Y: Data Repository Scopus 63. [Data set]. *Zenodo.* 2025.
[Publisher Full Text](#)
- Zahra SA, Liu W, Si S: **How digital technology promotes entrepreneurship in ecosystems.** *Technovation.* 2023; **119**: 102457.
[Publisher Full Text](#)
- Zavodna LS, Überwimmer M, Frankus E: **Barriers to the implementation of artificial intelligence in small and medium-sized enterprises: Pilot study.** *Journal of Economics and Management.* 2024; **46**: 331–352.
[Publisher Full Text](#)
- Žilka M, Kalender ZT, Lhota J, *et al.*: **Tools to support managerial decision-building competencies in data driven decision making in manufacturing SMEs.** *Procedia Computer Science.* 2024; **232**: 416–425.
[Publisher Full Text](#)

Open Peer Review

Current Peer Review Status:  

Version 1

Reviewer Report 29 December 2025

<https://doi.org/10.5256/f1000research.189110.r431707>

© 2025 Jain A. This is an open access peer review report distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.



Arjita Jain 

NCRD's Sterling Institute of Management Studies,, Navi Mumbai, India

This study provides strong evidence of AI's growing and transformative role in SMEs, particularly in innovation, decision-making, and competitive strategy. Drawing on a B-SLR, the authors map key themes, methodological patterns, and research gaps, offering a clear overview of current scholarship on AI adoption in SMEs.

The bibliometric analysis shows a sharp rise in academic interest after 2020, mirroring global moves toward digital transformation and Industry 4.0. This surge underscores AI's emergence as a strategic tool for SMEs seeking operational efficiency and value creation.

Overall, the study's identification of major thematic clusters—AI-driven innovation, entrepreneurial decision-making, and adoption barriers and enablers—highlights the complex and multidimensional nature of AI integration within the SME sector.

Are the rationale for, and objectives of, the Systematic Review clearly stated?

Yes

Are sufficient details of the methods and analysis provided to allow replication by others?

Yes

Is the statistical analysis and its interpretation appropriate?

Yes

Are the conclusions drawn adequately supported by the results presented in the review?

Yes

If this is a Living Systematic Review, is the 'living' method appropriate and is the search schedule clearly defined and justified? ('Living Systematic Review' or a variation of this term should be included in the title.)

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Reimagining Management Education, Education 4.0, SMEs, CBHRM, Employee Engagement, NEP 2020 etc

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Reviewer Report 08 November 2025

<https://doi.org/10.5256/f1000research.189110.r429211>

© 2025 kassa E. This is an open access peer review report distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.



Erstu kassa

Woldia University, Weldiya, Amhara, Ethiopia

The review topic is interesting and contemporary.

My comments and suggestions are;

1. Please remove the citation in table 3.
2. In the abstract section, remove numbers (1), (2), and (3).
3. The introduction section is well written.
4. Add a separate topic for the literature review.
5. The method section is good.

Are the rationale for, and objectives of, the Systematic Review clearly stated?

Yes

Are sufficient details of the methods and analysis provided to allow replication by others?

Yes

Is the statistical analysis and its interpretation appropriate?

Not applicable

Are the conclusions drawn adequately supported by the results presented in the review?

Yes

If this is a Living Systematic Review, is the 'living' method appropriate and is the search schedule clearly defined and justified? ('Living Systematic Review' or a variation of this term should be included in the title.)

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Knowledge management, digital transformation, strategic management

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

The benefits of publishing with F1000Research:

- Your article is published within days, with no editorial bias
- You can publish traditional articles, null/negative results, case reports, data notes and more
- The peer review process is transparent and collaborative
- Your article is indexed in PubMed after passing peer review
- Dedicated customer support at every stage

For pre-submission enquiries, contact research@f1000.com

F1000Research